

THE U.S. HOUSING AFFORDABILITY GAP

Capstone Final Report

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Overview

Homeownership has many positive financial and social benefits for the homeowner, the community, and the country as a whole, and it is still a life goal for roughly 61% of Americans. [1,2] Some benefits of owning a home include control over expenses, building wealth, and building credit to be better positioned to face future life events or obstacles. Since the 1981, the median U.S. home price has increased by approximately 118% and is currently at a historic high. By comparison, median household income increased only 22% over the same period, creating a housing affordability gap for lower and median-income households, and even for many two-income families. Fewer Americans buying homes means fewer of them will enjoy the benefits of home ownership while continuing to rent rather than building equity in an appreciating asset.

The goal of this project is to understand and present the drivers behind the home affordability gap and to a lesser extent, examine the drivers behind flat median income growth over the last 40-years.

Data Analysis & Computation

Data Sources

The initial phase of our analysis focused on exploring the residential real estate market and the many variables that affect it in order to determine and locate reputable data sources and market analysis. Ultimately, our analysis utilized a combination of data sets from government and real estate industry sources. The primary data sources used were the Federal Reserve Economic Data (FRED) [3] system and the U.S. Census Bureau [4], the data from which was also accessed through the FRED system. Secondary data and analysis came from Realtor.com research [5] and independent real estate market analysts, such as realestatewitch.com [6]

Data Wrangling & Cleaning

The primary variables utilized to investigate the housing affordability gap include home prices, mortgage interest rates, housing inventory, household income, U.S. population and inflation. While some data was available back to the 1960's, we focused on the period from 1981 to 2021. This forty-year period offered the most consistent historic data availability across the selected variables and covered five U.S. recessions, including the recent COVID recession.

The FRED system is highly reputable and provided the ability to customize our extracted data sets by time period and level of detail (e.g., from months or quarters to years) between 1981 - 2022 (figure 1).

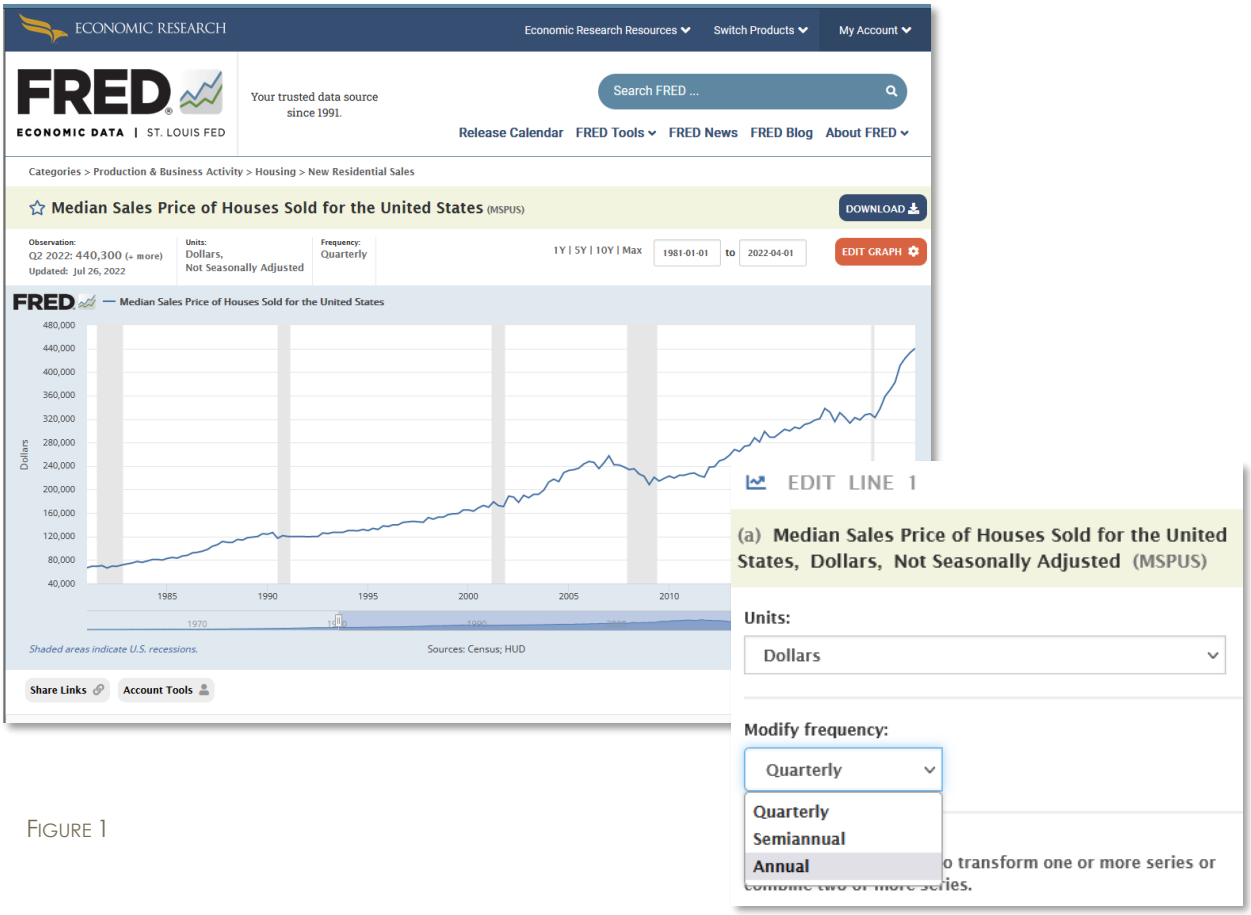
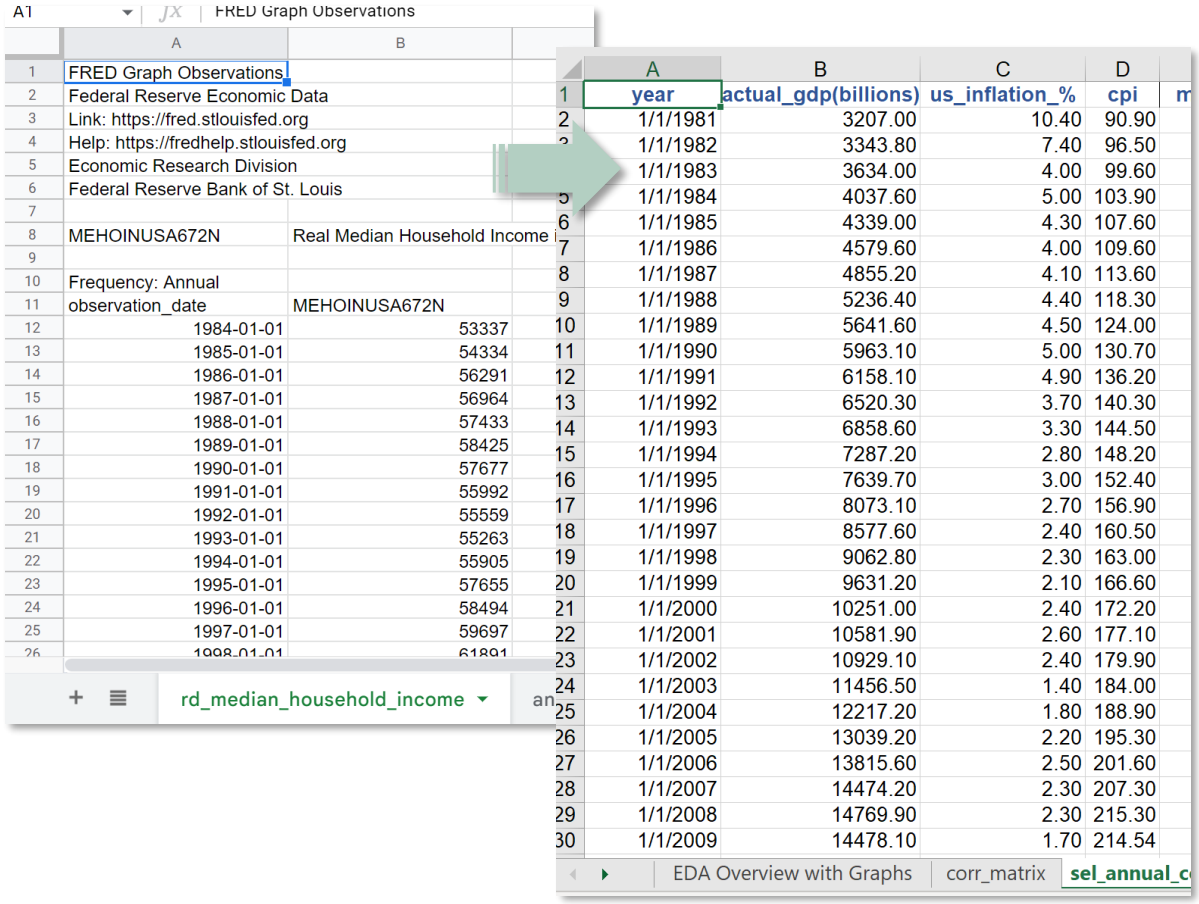


FIGURE 1

The raw data extracted from FRED were separated by variables that would eventually become columns. Our process began by downloading the raw data files in .csv or .xls format and then transforming the layout into a consistent tabular data structure (figure 2). For this process, we agreed on using cohesive titles and formatting. We made sure column titles were all descriptive, lowercase, contained no spaces, and remained short.

The next step was to begin calculating averages and rate columns and creating a consistent annualized date set by year for 1981 – 2022.



	A	B
1	FRED Graph Observations	
2	Federal Reserve Economic Data	
3	Link: https://fred.stlouisfed.org	
4	Help: https://fredhelp.stlouisfed.org	
5	Economic Research Division	
6	Federal Reserve Bank of St. Louis	
7		
8	MEHOINUSA672N	Real Median Household Income i
9		
10	Frequency: Annual	
11	observation_date	MEHOINUSA672N
12	1984-01-01	53337
13	1985-01-01	54334
14	1986-01-01	56291
15	1987-01-01	56964
16	1988-01-01	57433
17	1989-01-01	58425
18	1990-01-01	57677
19	1991-01-01	55992
20	1992-01-01	55559
21	1993-01-01	55263
22	1994-01-01	55905
23	1995-01-01	57655
24	1996-01-01	58494
25	1997-01-01	59697

rd_median_household_income

	A	B	C	D
1	year	actual_gdp(billions)	us_inflation_%	cpi
2	1/1/1981	3207.00	10.40	90.90
3	1/1/1982	3343.80	7.40	96.50
4	1/1/1983	3634.00	4.00	99.60
5	1/1/1984	4037.60	5.00	103.90
6	1/1/1985	4339.00	4.30	107.60
7	1/1/1986	4579.60	4.00	109.60
8	1/1/1987	4855.20	4.10	113.60
9	1/1/1988	5236.40	4.40	118.30
10	1/1/1989	5641.60	4.50	124.00
11	1/1/1990	5963.10	5.00	130.70
12	1/1/1991	6158.10	4.90	136.20
13	1/1/1992	6520.30	3.70	140.30
14	1/1/1993	6858.60	3.30	144.50
15	1/1/1994	7287.20	2.80	148.20
16	1/1/1995	7639.70	3.00	152.40
17	1/1/1996	8073.10	2.70	156.90
18	1/1/1997	8577.60	2.40	160.50
19	1/1/1998	9062.80	2.30	163.00
20	1/1/1999	9631.20	2.10	166.60
21	1/1/2000	10251.00	2.40	172.20
22	1/1/2001	10581.90	2.60	177.10
23	1/1/2002	10929.10	2.40	179.90
24	1/1/2003	11456.50	1.40	184.00
25	1/1/2004	12217.20	1.80	188.90
26	1/1/2005	13039.20	2.20	195.30
27	1/1/2006	13815.60	2.50	201.60
28	1/1/2007	14474.20	2.30	207.30
29	1/1/2008	14769.90	2.30	215.30
30	1/1/2009	14478.10	1.70	214.54

EDA Overview with Graphs | corr_matrix | sel_annual_c

FIGURE 2

The primary tools used were Excel PivotTables and formulas for new columns added. We combined annual correlated factors into one table by applying years as the rows. During this process, tables with years as columns had to be transposed in order to match the format of the consolidated datasets by using PivotTables or the "Text to Columns" feature in Excel.

Our cleaning process included organizing the worksheets and adding descriptive titles to streamline analyses in the future. The final step in the data cleaning process was formatting column titles and values suitable for use in Excel and Tableau, and finally incorporating the data into our EDA Excel workbook.

Exploratory Data Analysis

The residential real estate market is a broad and complex area of research with literally hundreds of variables affecting it simultaneously. Given the available time, budget (none), and finite resources with limited knowledge of the housing market, we had to narrow the scope of our analysis to focus on a handful of industry and macro-economic variables.

The EDA process began by becoming familiar with the numerous data sets to understand their content, units of measure and context relative to the other data and the housing market as a whole. Each data set was also examined for null values, of which there were very few – nulls were primarily due to incomplete or missing 2022 data. Because of its importance in the analysis, CPI data was extrapolated for 2022 using the YTD value to present the full year.

The next step was to create a correlation matrix across the variables to understand their relationships to one another with green representing the strongest *positive* correlations and red the strongest *negative* correlations (figure 3).

	actual_gdp(billions)	US_inflation_%	CPI	us_population(1000s)	median_houseold_income(USD)
actual_gdp(billions)	1				
US_inflation_%	-0.653032022	1			
CPI	0.991354032	-0.716138624	1		
us_population(1000s)	0.986560303	-0.734433054	0.997379003	1	
median_houseold_income(USD)	0.731720126	-0.466603945	0.715914058	0.709301829	1
number_of_households	0.988285298	-0.726736098	0.997173491	0.998315605	0.716864508
median_sold_home_price(USD)	0.990214848	-0.622035585	0.975324162	0.967613398	0.758316312
average_30-year_mortage_rate	-0.891434532	0.860766765	-0.931730726	-0.932456492	-0.664432347
annual_avg_months_supply_new_houses	-0.149266472	0.410688992	-0.167234373	-0.191417485	-0.159451859
annual_avg_new_homes_for_sale	-0.158944285	0.099538326	-0.176037121	-0.178070306	0.207851205
new_home_price_index(Laspeyres)	0.989094561	-0.625344688	0.974968394	0.967005627	0.779144855
annual_home_ownership_rate	0.219163224	-0.364248172	0.271094527	0.301229776	0.376270841
annual_rental_vacancy_rate	0.252936886	-0.630718377	0.343500813	0.369639443	0.21784382

FIGURE 3 – PARTIAL VIEW

Basic line graphs were created for single and multiple variables to help visualize time series trends and interactions between the data, keeping the previous correlation data in mind. Analysis also included calculating year-over-year percent change and beginning to ending period-to-period change from 1981 – 2022. Industry standard ratios were used, including home price-to-income, average new home months of supply, along with two we created:

1. *Rental-to-For Sale Housing*: based on the strong (0.92) correlation between number of homes for sale and number of housing units available for rent.
2. *House-to-Population*: measures the number residential housing units available per person in the U.S.

Lastly, U.S. recessions, particularly the 2008 housing market crash, were also evaluated for impact on the housing market and household incomes.

Understanding The Gap

Record High Home Prices

The first step towards understanding the home affordability gap was to investigate why median home prices rose significantly between 1981 and 2022. Four primary factors contributed to the increase: low mortgage interest rates, insufficient housing inventory, an increasing number of homebuyers primarily due to population growth and Federal legislation that relaxed mortgage lending standards (Figures 4 & 5).

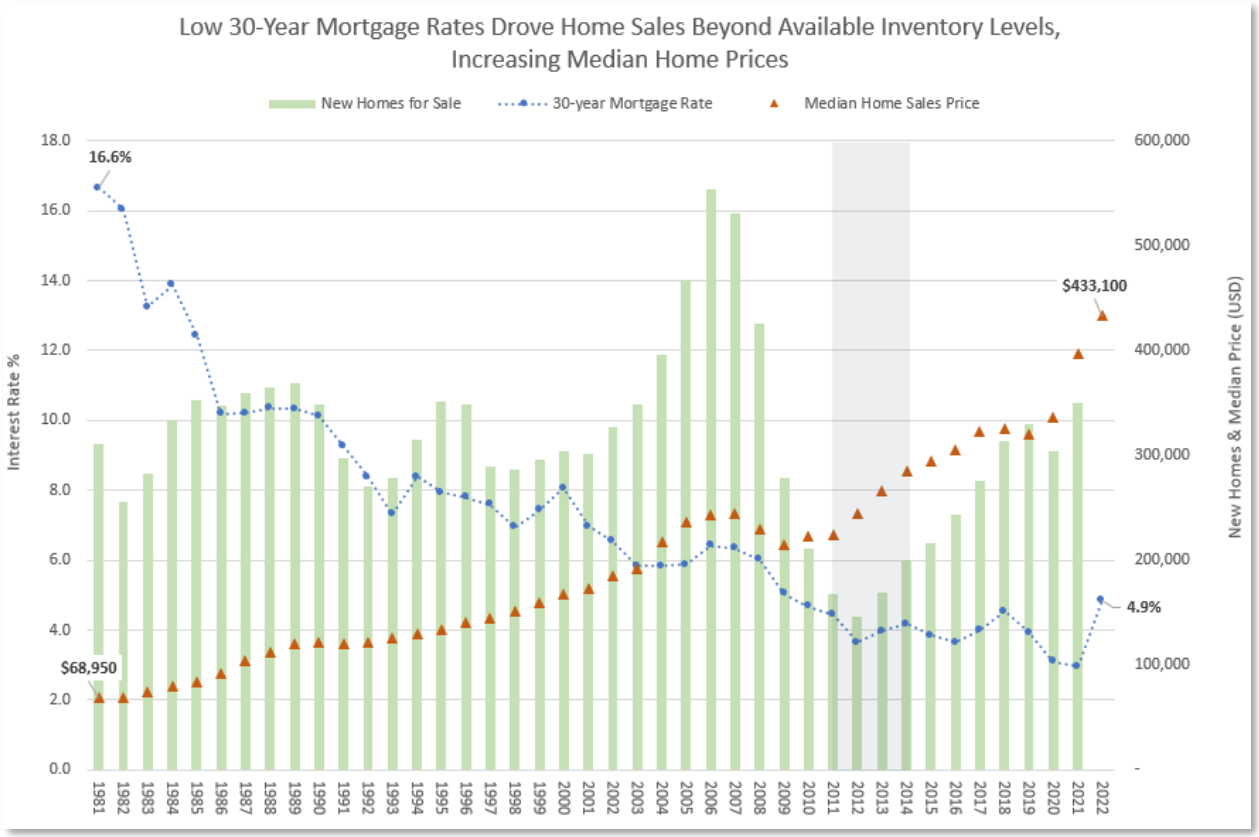


FIGURE 4

The average 30-year mortgage rate dropped from 16% in 1981 to a 40-year low of 3.0% in 2021. Low mortgage rates made houses more affordable relative to a given income and increased buyer demand for homes. At the same time, new home inventory growth slowed after 1990, with few exceptions, before soaring in the years leading up to

the 2008 crash, and then collapsing to an all-time low in 2012. U.S. population grew steadily from 1981- 2006 adding 68 million people, a 40% increase (Figure 5).

Together, these factors increased housing demand beyond available inventory levels and forced prices upward. The greatest impact occurred between 2011 – 2104 when new housing inventory was at its lowest level since 1981 and interest rates continued to fall to 4% in 2012, causing home demand to outstrip available inventory, steeply increasing median home prices to then-record levels (Figure 4, grey area).

Anemic Household Income Growth

From 1981 to 2022 (YTD), median U.S. household income grew by only 22% while median home prices increased by 118% over the same period. This is the other half of the affordability gap. Low mortgage interest rates helped offset minimal income growth by making more expensive homes more affordable without a parallel increase in median income (figure 5).

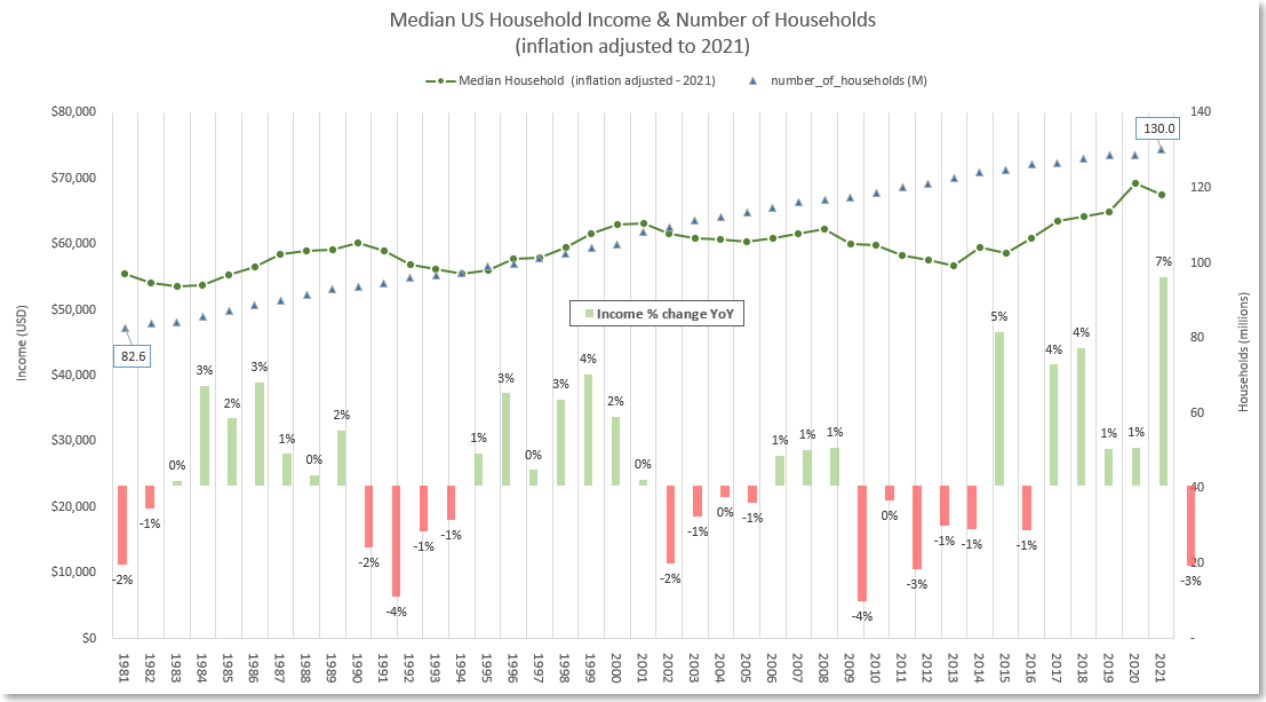


FIGURE 5

Inflation was and is a major contributor to historically anemic household income growth. As measured by the Consumer Price Index (CPI), inflation grew significantly between 1981 – 2021 with the CPI increasing from 91 to 290, respectively. This

represents a significant erosion in household purchasing power and contributed to the affordability gap (Figure 6)

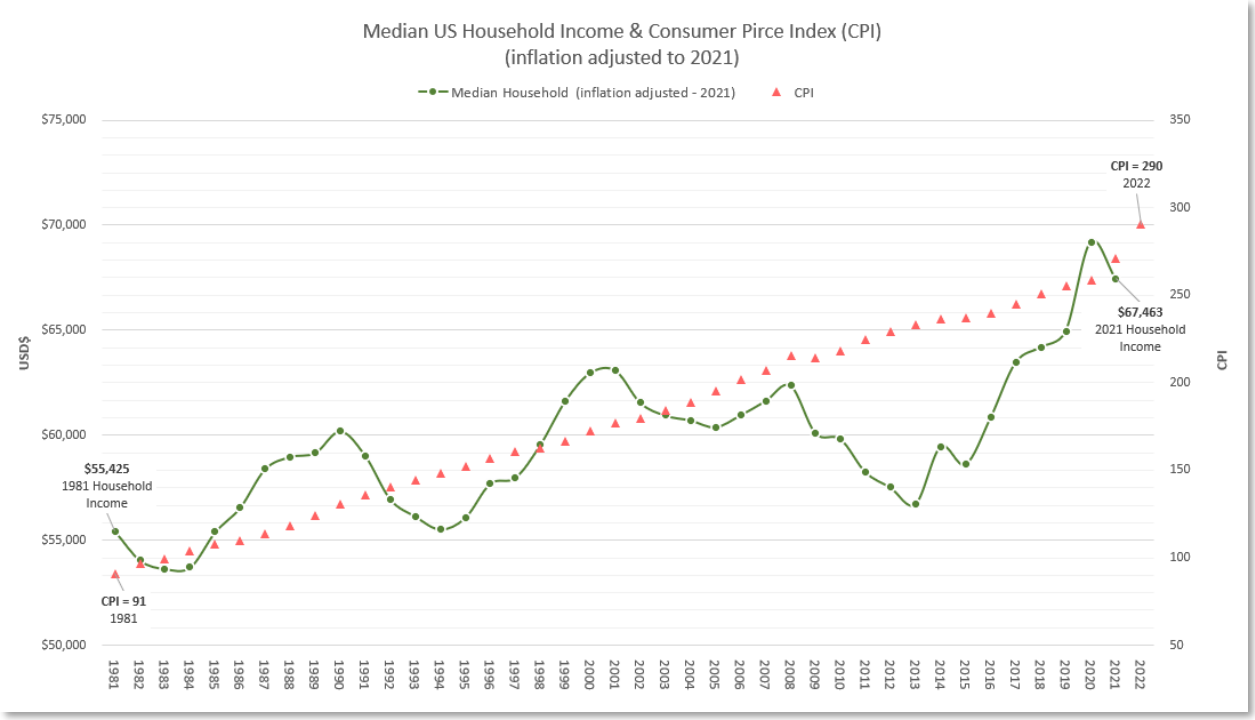


FIGURE 6

The COVID Effect

Beginning in 2020, the COVID pandemic had a cataclysmic impact on the U.S. economy, housing market, household incomes and the American people. Median home prices shot upwards from previous historic highs while median household income dropped as unemployment hit a post-depression record 14% [6] due to thousands of small businesses closing while others laid off employees (Figures 4 & 6). Inflation also soared to 9% this year, the highest rate since the early 1970's (Figure 6, CPI).

After 2022, home buying surged as people moved away from densely populated areas and renal units in favor of single-family homes, particularly those located in more rural areas.

Since 2020, investors also had a significant impact on the housing market by purchasing nearly 25% of all single-family homes sold in 2021 [7]. In many cases, investors competed directly against individual home buyers, out bidding them and in the process, drove up median home prices.

In addition, the FED has increased interest rates significantly since last fall, driving the average 30-year mortgage rate to 7%, highest since 2001.

As a result of these factors, the home affordability gap is at an all-time high, with the home price-to-income ratio reaching 5.88 in 2021, against a recommended maximum value of 2.5, which was last seen in 1997 (Figure 7). What this means is homebuyers with a minimum down payment – typically 20% to avoid the additional expense of Private Mortgage Insurance (PMI) – could have a mortgage that is approximately five times their annual income. This sets a dangerous precedent for highly leveraged homebuyers that may find themselves owing more than the market value of their home if the residential real estate market makes a significant price correction in the next few years.

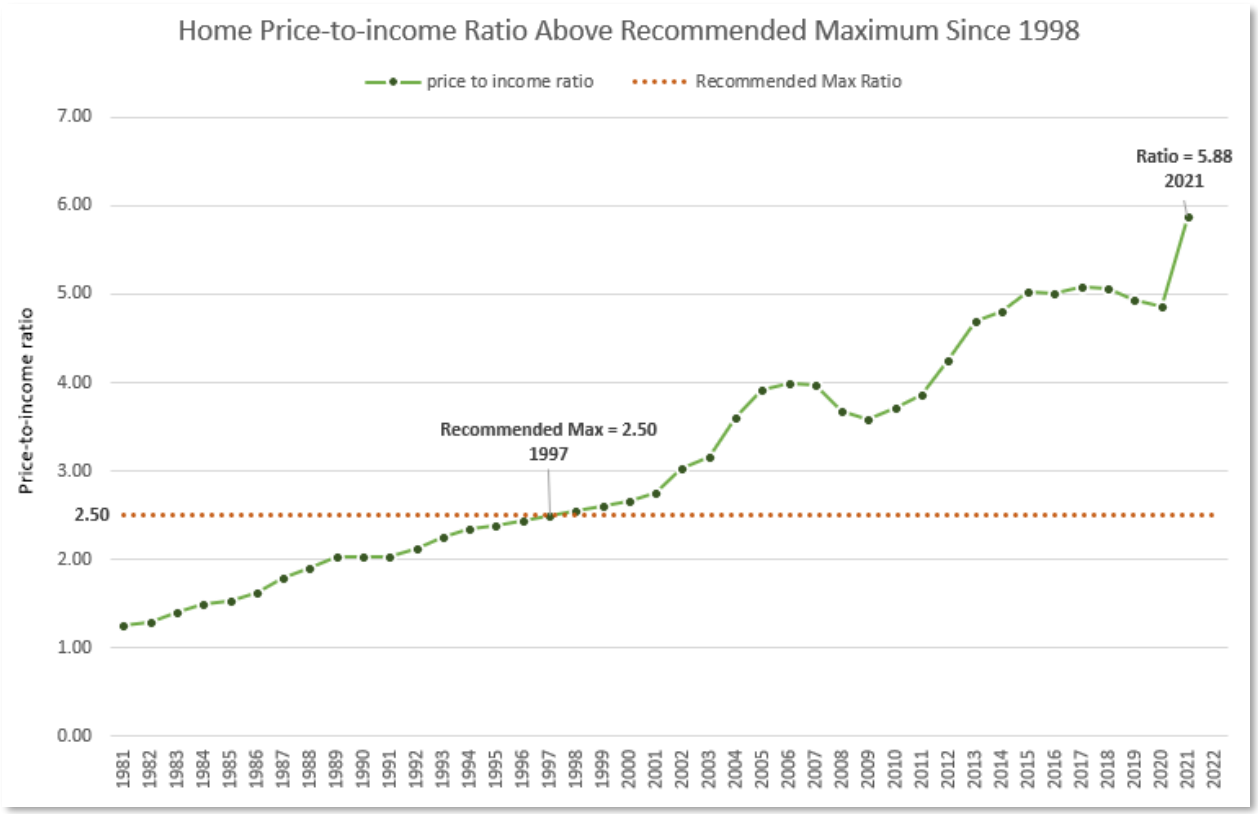


FIGURE 7

Dashboard

Link

The dashboard (Figure 8) was created to demonstrate not only the exponential growth of the housing market, but also highlight the massive difference when compared to median household income across the United States. We also took the opportunity to highlight what we believe to be one of the key contributing factors as to why houses were still being bought despite this gap, which would be low interest rates. We observed that while house prices went up, the interest rates went down.

Additionally, we proceed to show the different types of finance types used to purchase houses. We saw a spike within all the loan types, and we also saw a spike in 2021 cash purchases. In more recent years, we saw a record high when it came to investor bought homes. What is interesting is that it seems investors also used some type of conventional loan when purchasing these houses based on the number of cash bought houses versus total number of homes bought by investors.

Finally, we wanted our audience to see the 2021 job market sectors. We provided the top five hiring job markets, in hopes of helping our audience make informed education and career choice decisions, in order to become future homeowners.

Dashboard Components

- Median Sold Home Price, Median Household Income and Average Interest Rate
    - Section: Top Left
    - Graph Type: Line
- Last 10 Years Home Sales Finance Categories
    - Section: Bottom
    - Graph Type: Bar Graph
- Top 10 States for Investor Activity
    - Section: Top Right
    - Graph Type: Map
- Post COVID-19 Pandemic Hiring Sectors
    - Section: Mid Right

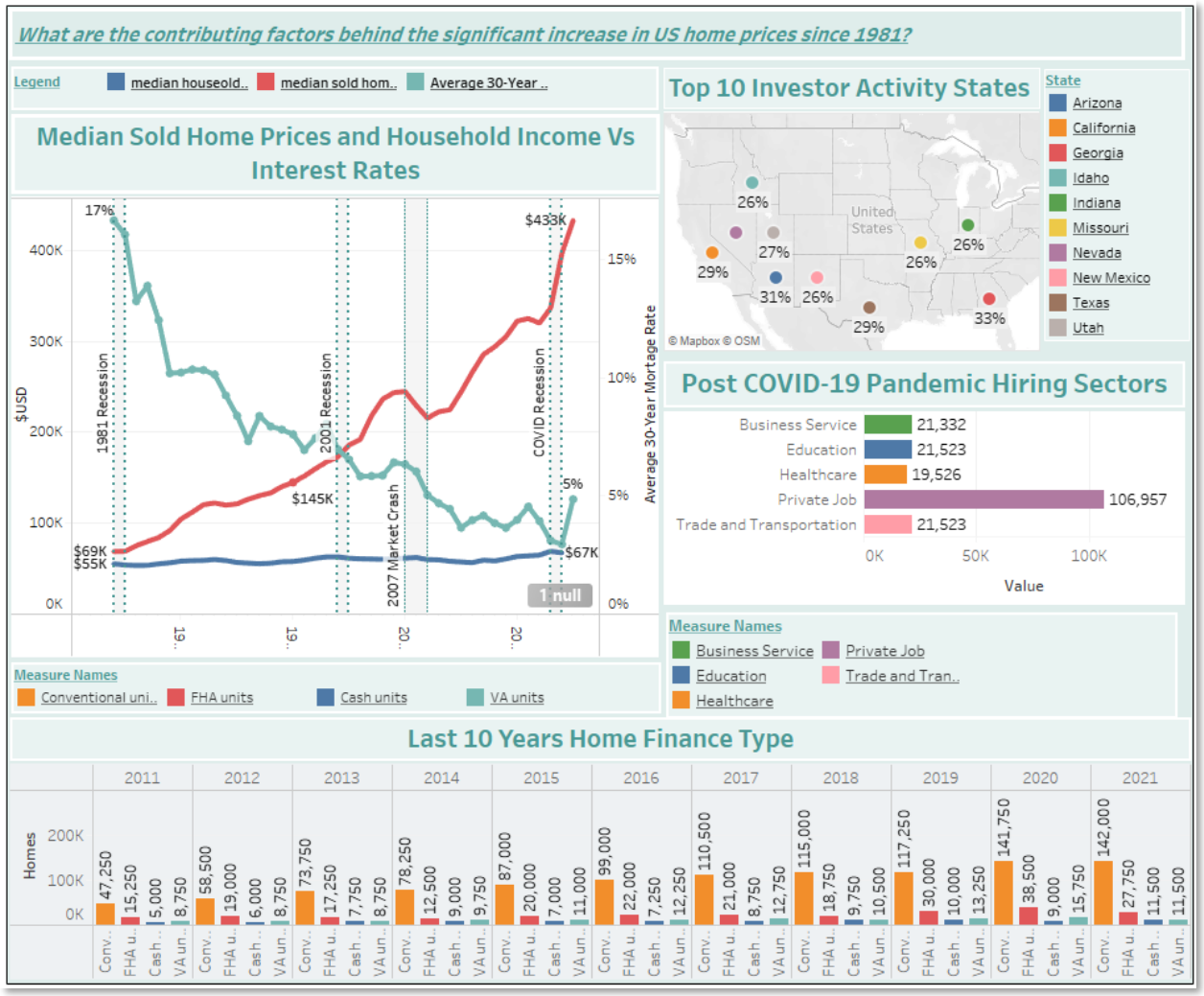


FIGURE 8

Tools Utilized

- Zoom
- Slack
- Excel
- Tableau
- Google Docs & Sheets
- Coffee. Lots of coffee.

Final Observations

The current housing market is in an untenable state due to the compounding effects of interest rates, insufficient housing inventory, ever increasing population and unchecked immigration, anemic household income, record high inflation and the likelihood of increased taxes at both the Federal and State level.

As more and more Americans cannot bridge the home affordability gap due to prohibitively high home prices and now, high mortgage rates, they have no alternative but to rent, which has contributed to rents increasing to historically high levels, making it even more challenging for them to save money for a down payment on a home.

For many years, Government policies have had a deleterious effect on residential housing prices, inflation, and household income, which were exacerbated during and after the COVID pandemic. This combined with investors now buying significant numbers of single-family homes puts additional pressure on an already strained housing inventory and continues to push home prices higher.

To close the housing affordability gap for median income families, household incomes must increase significantly while inflation, which is primarily driven by Federal and state government economic policies, must be reduced. New home inventory needs to increase to meet existing and future demand. The new home construction cost index (figure 9) is at an historic high of 177 against the base year value of 100 in 1997. This represents a 77% increase in new home construction costs. Reducing home construction costs through lower taxes and material costs should stimulate additional new home construction.

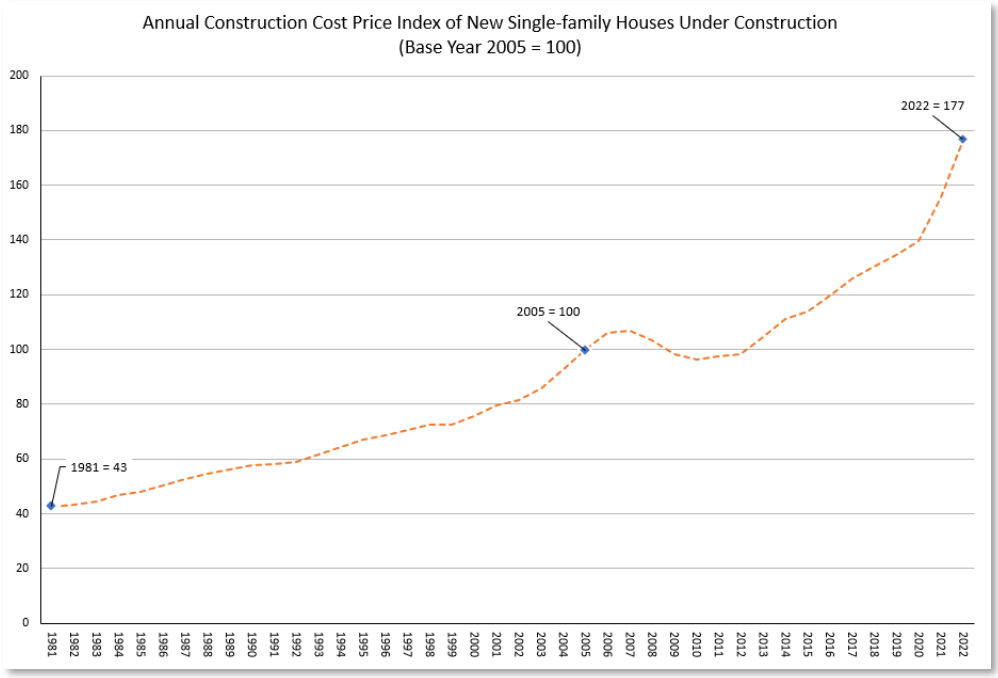


FIGURE 9

Sources

[1] National Association of Realtors Research

https://www.nar.realtor/research-and-statistics/housing-statistics/existing-home-sales

[2] "Why Home Ownership is Good for Everyone", Discover.com, https://www.discover.com/home-loans/articles/why-home-ownership-is-good-for-everyone

[3] Economic Research, Federal Reserve Bank of St. Louis

https://fred.stlouisfed.org/

[4] United States Census Bureau

https://www.census.gov/

[5] Realtor.com Research

https://www.realtor.com/research/

[6] FRED Unemployment Rate, percent, not seasonally adjusted.

https://fred.stlouisfed.org/series/U2RATE#0

[7] Pew Charitable Trusts, "*Investors Bought a Quarter of Homes Sold Last Year, Driving Up Rents*",

https://www.pewtrusts.org/en/research-and-analysis/blogs/stateline/2022/07/22/investors-bought-a-quarter-of-homes-sold-last-year-driving-up-rents#:~:text=Investors%20bought%2024%25%20of%20all,months%20of%202022%20to%2022%25